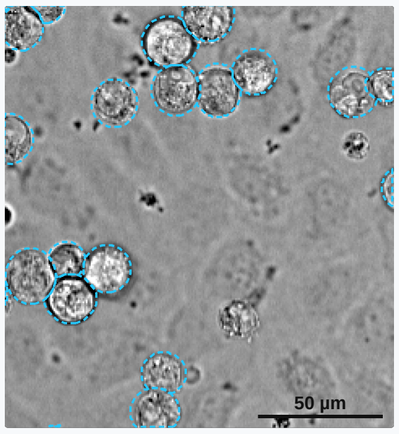
## a When a pre-trained model performs poorly, find the mismatch first

model — manual annotation

### 1 Check the normalisation

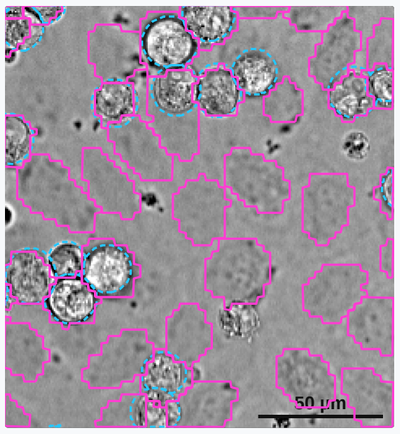
tile: forgot to normalise




| F1 | |
|---|---|
| **forgot to normalise** | **0.00** |
| percentile (default) | 0.78 |
| min–max | 0.81 |
| histogram equalisation | 0.55 |
| CLAHE | 0.60 |

### 2 Check the object scale

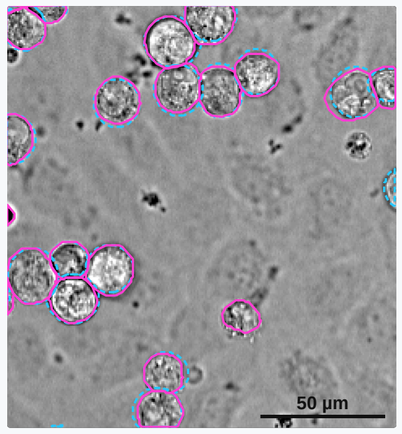
tile: input downsampled 4×




| F1 | |
|---|---|
| **downsampled 4×** | **0.33** |
| downsampled 2× | 0.40 |
| native (default) | 0.78 |
| upsampled 2× | 0.72 |

### 3 Check the post-processing

tile: low-contrast objects removed




| F1 | |
|---|---|
| none (default) | 0.78 |
| stricter probability threshold | 0.79 |
| looser flow threshold | 0.75 |
| **low-contrast objects removed** | **0.92** |

**4 If nothing works, fine-tune or train**

## b Predicted images: what will you measure?

**Used as an intermediate step**

visualise · segment · detect · register

→ **Validate the downstream output**

e.g. segment on the predicted image, compare with annotations of acquired images

**Used as the measurement itself**

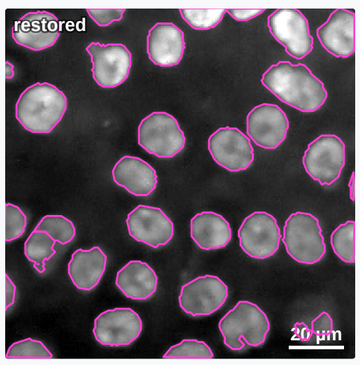
intensity · abundance · localisation

→ **Validate the measurement against acquired data**

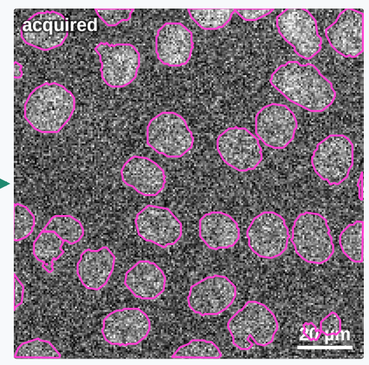
e.g. predicted vs acquired fluorescence, in independent paired samples

**Label predicted images as predicted; do not describe predicted fluorescence as measured.**

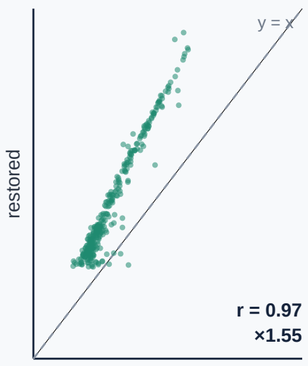
## c Denoising: measure intensity in the acquired image




mask from here

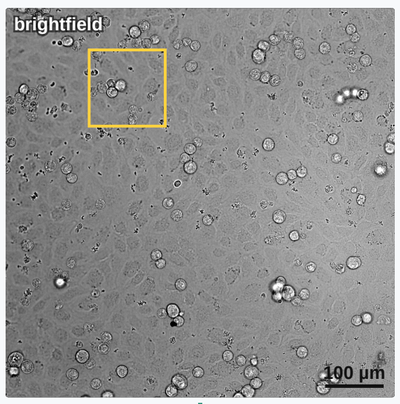



measure here



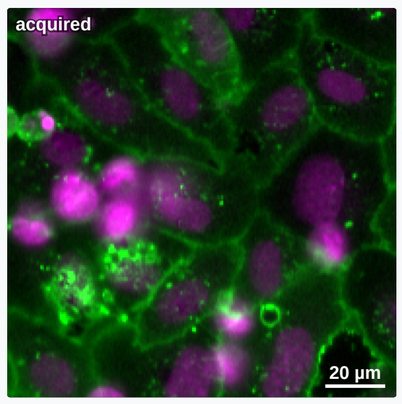

## d Artificial labelling: validate what you will measure

model — manual, annotator 1 — manual, annotator 2



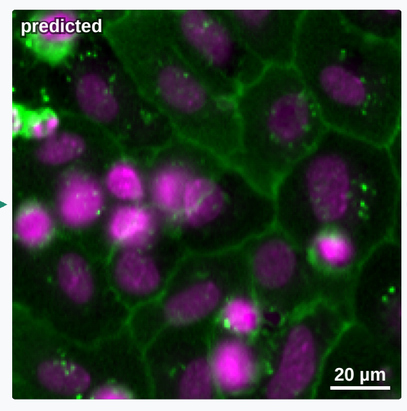

DL model: G → D (pix2pix)

DAPI + PECAM-1

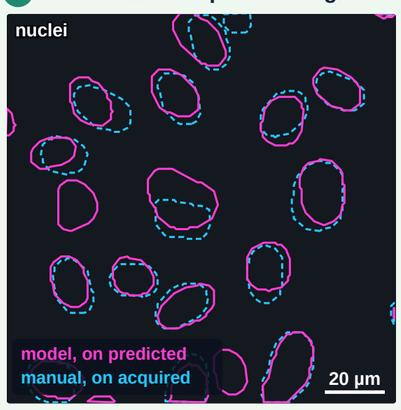



manual annotation

model segmentation

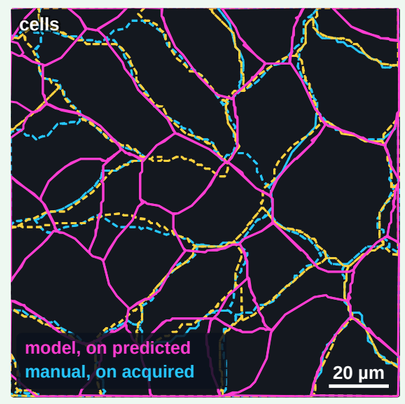



**Intermediate step: to aid segmentation**




**Nuclei · F1**

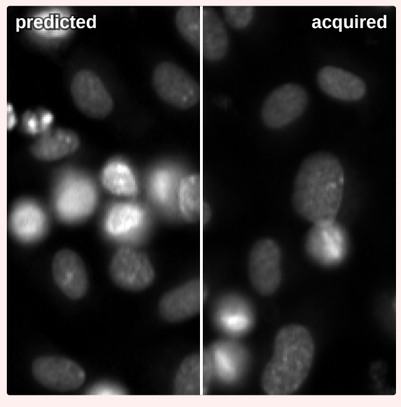
| model vs manual nuclei | 0.83 |
|---|---|

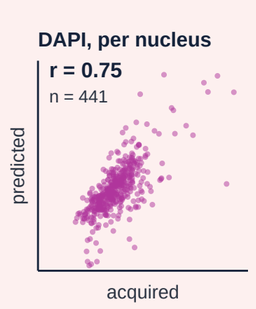


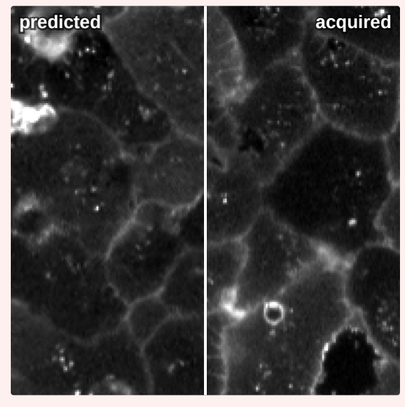

**Cells · F1**

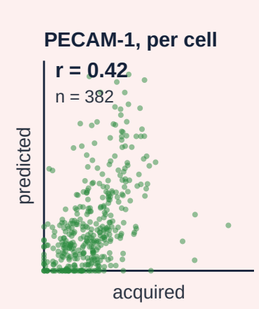
| annotator 1 vs 2 | 0.90 |
|---|---|
| model vs annotator 1 | 0.78 |
| model vs annotator 2 | 0.75 |

**Measurement itself: intensity**



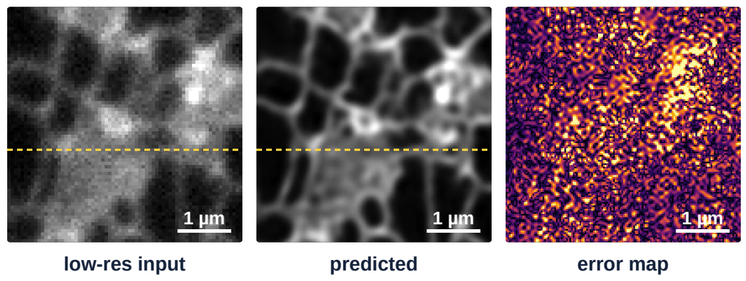




## e Super-resolution: check against the input and, where possible, higher-resolution data

**Consistency with the diffraction-limited input (SQUIRREL)**

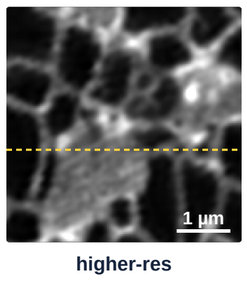


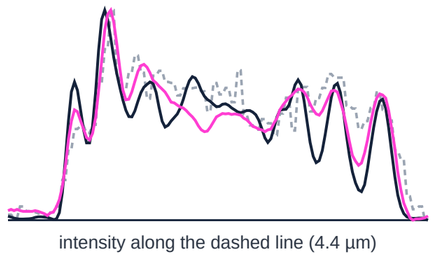

**RSP with the input**

| predicted | 0.98 |
|---|---|
| higher-res | 0.93 |

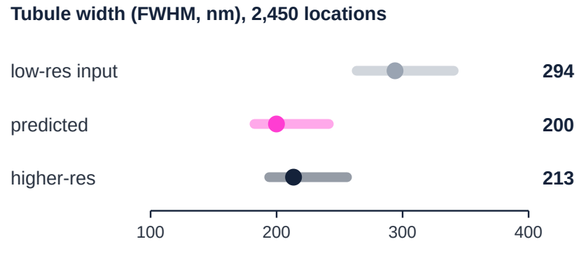
0.90 · 0.95 · 1.00

**Comparison with higher-resolution data: test the feature you will measure**




**Tubule width (FWHM, nm), 2,450 locations**




**predicted − higher−res, same location: −13 nm (IQR −32 to +8)**